

IoT project completion roadmap

What is already available: Title, description, components list (finalized)

What is the ultimate goal: To demonstrate the proof of concept (lab scale model)

Week 1: To complete design

Goal by end of Week-1

A complete system definition along with circuit design and if possible simulation studies that can be built immediately once parts arrive.

Day 1: Problem definition, measurable requirements (1-page)

Deliverables

- 1. Problem statement (5 lines), target user, use scenario.**
- 2. Target metrics (minimum 6) – measurable**
Examples: Accuracy, threshold detection time, battery backup hours, alert latency, communication range, max load, false alarm rate.
- 3. Constraints (minimum 6) – cost cap, size, mounting, weather/IP rating, safety.**

Day 2: Architecture, block diagram, data flow

Deliverables

1. Block diagram (power → sensing → processing → comms → alert).
2. Data flow diagram (sensor sampling → filtering → feature → decision → logging → transmission).
3. Decision logic draft: what triggers green/amber/red; what triggers buzzer; what triggers transmission.

Day 3: Circuit schematic (complete, pin-mapped)

Deliverables

1. Schematic in KiCad/EasyEDA (or neat hand schematic with pin table).
2. Pin mapping table (MCU pin $\rightarrow$ module pin $\rightarrow$ function).
3. Power budget table (each module current, total, expected battery life).

Check: Schematic must include protections (reverse polarity, charger protection, regulator, fuse/TVS if needed).

Day 4: Firmware skeleton, interface contracts (without hardware)

Deliverables

1. Firmware folder with:

- sensor read functions (mocked values)
- filtering function
- decision/threshold function
- logging format (CSV/JSON)
- comms packet structure

2. Packet format spec (1 page): fields, units, sampling rate, checksum if any.

Check: Must compile (Arduino IDE/ PlatformIO) even with mocked sensors.

Day 5: Virtual prototype / simulation proof

Pick any one method (must show evidence):

- Wokwi/Proteus simulation (MCU logic with mocked I/O)
- MATLAB/Simulink logic validation
- Python simulation of signal $\rightarrow$ filter $\rightarrow$ threshold classifier
- LTspice for analog front-end (if applicable)

Deliverables

1. Simulation results: at least 3 scenarios

- normal/safe
- warning
- overload/fault

2. Plots / logs showing decision transitions.

Week-1 Gate (must submit)

- Requirements sheet
- Block + data flow
- Schematic + pin map
- Firmware skeleton compiles
- Simulation evidence

Week 2: Novelty, inventive step preparation for patent

Goal by end of Week-2

A patent-ready novelty & inventive step package along with build plan and test plan and “prototype sprint plan” for Week-4.

Day 6: Prior art search (structured, not random)

Deliverables

1. Prior art table (minimum 12 entries)

Columns: Title/Link, Type (paper/patent/product), What it does, What it misses, Your differentiation.

2. Closest 3 prior arts flagged (the “threats”).

Check: At least 4 patents + 4 products + 4 papers/articles (or equivalent mix).

Day 7: Novelty carving (feature list , differentiation map)

Deliverables

1.Feature inventory (minimum 15 features)

grouped as:

- sensing
- algorithms
- mechanical integration/mounting
- safety/alerts
- communication/logging

2.Differentiation map: “Feature vs Prior-art gap” (what is new and why it matters).

Check: Must include at least 3 claim-worthy features (not just “uses ESP32”).

Day 8: Inventive step framing (problem–solution–effect)

Deliverables

1. Inventive step sheet (1–2 pages):

- Technical problem
- Proposed solution
- Technical effect (quantified where possible)
- Why non-obvious vs closest prior art

2. Failure modes & safety logic (minimum 8 failure modes + mitigation)

Check: Must show “technical effect” (not only “cost-effective”).

Day 9, 10: Practice with already available components

Week 3: Prototype building